

## 1. BASIC LINUX COMMANDS

| Command      | Sample Usage         | Purpose                                         |
|--------------|----------------------|-------------------------------------------------|
| <b>pwd</b>   | pwd                  | Shows your current directory path.              |
| <b>ls</b>    | ls -la               | Lists files and folders, including hidden ones. |
| <b>cd</b>    | cd Documents         | Moves to another directory.                     |
| <b>mkdir</b> | mkdir project        | Creates a new folder.                           |
| <b>touch</b> | touch file.txt       | Creates an empty file.                          |
| <b>cp</b>    | cp a.txt b.txt       | Copies a file or folder.                        |
| <b>mv</b>    | mv old.txt new.txt   | Moves or renames a file.                        |
| <b>rm</b>    | rm file.txt          | Deletes a file.                                 |
| <b>cat</b>   | cat file.txt         | Displays file contents.                         |
| <b>grep</b>  | grep "error" log.txt | Searches for matching text.                     |
| <b>head</b>  | head -n 5 file.txt   | Shows the first lines of a file.                |
| <b>tail</b>  | tail -n 5 file.txt   | Shows the last lines of a file.                 |

## 2. BASIC GIT COMMANDS

| Command             | Sample Usage           | Description                                   | Common Options / Flags        |
|---------------------|------------------------|-----------------------------------------------|-------------------------------|
| <b>git init</b>     | git init               | Starts a new Git repository.                  | -q, -b main                   |
| <b>git restore</b>  | git restore file.txt   | Reverts a file to its tracked version.        | --staged, --source, -p        |
| <b>git status</b>   | git status             | Shows changed, staged, and untracked files.   | -s, -b                        |
| <b>git branch</b>   | git branch feature     | Creates or lists branches.                    | -a, -d, -m                    |
| <b>git add</b>      | git add .              | Stages changes for the next commit.           | ., -A, -p                     |
| <b>git checkout</b> | git checkout main      | Switches branches or restores files.          | -b, --                        |
| <b>git switch</b>   | git switch feature     | Changes to another branch.                    | -c, -d                        |
| <b>git commit</b>   | git commit -m "Update" | Saves staged changes as a commit.             | -m, -a, --amend               |
| <b>git merge</b>    | git merge feature      | Combines another branch into the current one. | --no-ff, --abort              |
| <b>git log</b>      | git log --oneline      | Displays commit history.                      | --oneline, --graph, -n        |
| <b>git tag</b>      | git tag v1.0           | Labels a specific commit, often for releases. | -a, -d, -l                    |
| <b>git diff</b>     | git diff               | Shows unstaged changes.                       | --cached, --stat, --name-only |
| <b>git clone</b>    | git clone <URL>        | Downloads a remote repository.                | --depth, -b                   |

## GIT THREE-TREE ARCHITECTURE→

"What I am editing"→ git add . → "What I will commit"→ git commit -m "message" → "What I committed → git push

## REFERENCES

Git Documentation — [git-scm.com/docs](https://git-scm.com/docs)

Pro Git — Three-Tree Architecture — Git Tools: Reset Demystified

GNU Coreutils — Linux Commands — GNU Coreutils Manual

"I have not used any AI or AI-assisted technologies to prepare this work."